

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277680

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

IWATA; Tomonari et al.

GRIP DETECTION DEVICE FOR STEERING WHEEL

Abstract

A grip detection device for a steering wheel includes: a first electrode arranged at a rim; a second electrode arranged at the rim; and a control circuit, in which the first electrode is wider in projected area than the second electrode, the control circuit allows switching between: a first mode in which the first electrode serves as a sensor electrode and the second electrode serves as a shield electrode; and a second mode in which the first electrode serves as the shield electrode and the second electrode serves as the sensor electrode, and the control circuit further: obtains a first-mode capacitance detected by the first electrode serving as the sensor electrode in the first mode and a second-mode capacitance detected by the second electrode serving as the sensor electrode in the second mode; and detects a grip on the steering wheel based on the first-mode capacitance and the second-mode capacitance.

Inventors: IWATA; Tomonari (Osaka, JP), NAITOU; Hiroshi (Osaka, JP), SAWADA; Takateru (Osaka, JP)

Applicant: Panasonic Automotive Systems Co., Ltd. (Kanagawa, JP)

Family ID: 96881090

Appl. No.: 19/049649

Filed: February 10, 2025

Foreign Application Priority Data

JP	2024-030176	Feb. 29, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: G01D5/24 (20060101); B62D1/04 (20060101); H05B3/06 (20060101)

U.S. Cl.:

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application is based on and claims priority of Japanese Patent Application No. 2024-030176 filed on Feb. 29, 2024.

FIELD

[0002] The present disclosure relates to a grip detection device for a steering wheel.

BACKGROUND

[0003] PTL 1 discloses a sensor pad that includes a pad substrate, a sensor electrode, and a shield electrode. The sensor electrode and the shield electrode are spaced apart from and parallel to each other on a first surface of the pad substrate. The shield electrode is electrically coupled to a voltage source to create a capacitance between the shield electrode and the sensor electrode, and the sensor electrode can detect a change in the capacitance. The shield electrode can also heat the surface of a vehicle part. The sensor electrode can sense presence of an occupant's hands or body adjacent the steering wheel or seat assembly.

CITATION LIST

Patent Literature

[0004] PTL 1: International Publication No. WO2019/022765

SUMMARY

[0005] However, the sensor pad according to PTL 1 can be improved upon.

[0006] In view of this, the present disclosure provides a grip detection device for a steering wheel that is capable of improving upon the above related art.

[0007] A grip detection device for a steering wheel according to one aspect of the present disclosure includes: a first electrode arranged at a rim of the steering wheel; a second electrode arranged at the rim and in a more inner layer of the steering wheel than the first electrode; and a control circuit, in which the first electrode is wider in projected area than the second electrode, the control circuit allows switching between: a first mode in which the first electrode serves as a sensor electrode and the second electrode serves as a shield electrode; and a second mode in which the first electrode serves as the shield electrode and the second electrode serves as the sensor electrode, and the control circuit further: obtains a first-mode capacitance detected by the first electrode serving as the sensor electrode in the first mode and a second-mode capacitance detected by the second electrode serving as the sensor electrode in the second mode; and detects a grip on the steering wheel based on the first-mode capacitance and the second-mode capacitance.

[0008] The grip detection device for the steering wheel according to the present disclosure is capable of improving upon the above related art.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] These and other advantages and features of the present disclosure will become apparent from the following description thereof taken in conjunction with the accompanying drawings that illustrate a specific embodiment of the present disclosure.

[0010] FIG. 1 illustrates a passenger compartment of a vehicle in which a steering wheel according to Embodiment 1 is disposed.

[0011] FIG. 2 is a perspective cross-sectional view of the steering wheel taken at line II-II in FIG.

1.

[0012] FIG. 3 is a circuit configuration of a grip detection device for a steering wheel according to Embodiment 1.

[0013] FIG. 4 is a diagram illustrating capacitances in cases where a first mode, a second mode, and a third mode are performed when a user's finger is touching the steering wheel.

[0014] FIG. 5 illustrates cases where each mode is driven in a time sharing manner.

DESCRIPTION OF EMBODIMENT

[0015] Hereinafter, an embodiment is described in detail with reference to the drawings.

[0016] The embodiment described below presents a general or specific example. The numerical values, shapes, materials, elements, the arrangement and connection of the elements, etc., described in the following embodiment are mere examples, and therefore are not intended to limit the present disclosure. Accordingly, among elements in the following embodiment, those not appearing in any of the independent claims are described as optional elements.

[0017] The figures are schematic diagrams and are not necessarily precise illustrations. In the figures, like elements share like reference numbers.

[0018] In the following embodiments, expressions such as “T-shaped” are used. For example, “T-shaped” not only means exactly T-shaped, but also includes substantially T-shaped, i.e., includes an error of approximately several percent. In addition, “T-shaped” means T-shaped to an extent that the advantageous effects of the present disclosure can be achieved. The same applies to other expressions using “shaped”.

Embodiment 1

[0019] Firstly, the configuration of grip detection device for steering wheel 3 is described with reference to FIG. 1 and FIG. 2.

[0020] FIG. 1 illustrates a passenger compartment of vehicle 4 in which steering wheel 1 according to Embodiment 1 is disposed. FIG. 2 is a perspective cross-sectional view of steering wheel 1 taken at line II-II in FIG. 1.

[0021] As illustrated in FIG. 1 and FIG. 2, steering wheel 1 imparts a steering angle to, for example, the steering of vehicle 4.

[0022] Steering wheel 1 includes annular-shaped rim 2 and grip detection device for steering wheel 3. Rim 2 is integrally formed with a T-shaped spoke arranged on the inner circumferential surface of rim 2. Rim 2 is a grip part for gripping in driver's hand, and is ring-shaped.

[0023] Rim 2 includes core metal 2a which is an annular-shaped metal core, and resin layer 2b which covers core metal 2a. First electrode 21, second electrode 22, and base material 23 of grip detection device for steering wheel 3 are wrapped around rim 2. First electrode 21, second electrode 22, and base material 23 is described later. The outer circumference of first electrode 21 wrapped around rim 2 of grip detection device for steering wheel 3 is covered with steering cover 11 made of leather, a wood, a resin, etc. Steering cover 11 is adhered and fixed to the outer circumferential surface of first electrode 21 of grip detection device for steering wheel 3 with an adhesive, doubled-sided tape, or the like. Resin layer 2b is made of, for example, a urethane resin material such as foamed polyurethane.

[0024] Grip detection device for steering wheel 3 detects a touch of user's (person's) hand on steering wheel 1. The touch not only means a direct touch of user's hand on steering cover 11 of steering wheel 1, but also includes the state away from steering cover 11 as long as grip detection device for steering wheel 3 can sense user's hand. It is to be noted that grip detection device for steering wheel 3 can detect a touch even when, for example, user's arm or foot as well as hand touches steering wheel 1. In the present embodiment, the case where user's hand touches steering wheel 1 is described.

[0025] Grip detection device for steering wheel 3 is, for example, a capacitance-type proximity sensor, and is also a grip sensor that detects a grip with the hand of a user in vehicle 4 with steering wheel 1. Specifically, grip detection device for steering wheel 3 senses a change in capacitance

between the user's hand and first electrode **21** of grip detection device for steering wheel **3**, thereby detecting whether the user's hand is touching steering wheel **1**. When the user's hand is away from steering wheel **1**, grip detection device for steering wheel **3** detects the capacitance between vehicle **4** and first electrode **21**. However, the distance between vehicle **4** and first electrode **21** is extremely larger than the distance between first electrode **21** and the user's hand when the user's hand comes close to or is touching steering cover **11**, and thus the capacitance between vehicle **4** and first electrode **21** is minimal. Moreover, when the user's hand comes close to or is touching steering cover **11**, the capacitance changes since the user's hand intervenes between first electrode **21** and the vehicle body. When the sensed capacitance is greater than or equal to a defined value, it can be determined that the user's hand is touching or gripping steering wheel **1**.

[0026] Grip detection device for steering wheel **3** includes first electrode **21**, base material **23**, sewing thread **24**, second electrode **22**, control circuit **40**, and power supply circuit **40c**.

[0027] First electrode **21** is arranged around rim **2** of steering wheel **1**. Specifically, first electrode **21** is arranged on the inner surface of steering cover **11** and the outer surface of base material **23**. In other words, first electrode **21** is sandwiched between steering cover **11** and base material **23**. The outer surface of base material **23** faces steering cover **11**. First electrode **21** is fixed to base material **23** by adhering to the inner surface of steering cover **11** with an adhesive layer (not illustrated), and to the outer surface of base material **23** with an adhesive layer (not illustrated). First electrode **21** is connected to control circuit **40**.

[0028] It is to be noted that only one first electrode **21** may be arranged on the outer surface of base material **23**, or multiple first electrodes **21** may be arranged on the outer surface of base material **23**.

[0029] First electrode **21** is wider in projected area than second electrode **22**. As described later, second electrode **22** is a heater wire made of a conductive wire, and thus first electrode **21** is wider than second electrode **22** in the projected area when first electrode **21** and second electrode **22** are overlapped and viewed.

[0030] First electrode **21** is, for example, a solid electrode with a planar structure of a conductor, or a conductive fabric manufactured by applying plating to a non-metallic fiber. The non-metallic fiber is polyethylene terephthalate (PET) or the like. The plating is, for example, copper plating and nickel plating layered on the copper plating.

[0031] Base material **23** is a non-woven fabric of an elastic, flexible, and ductile material formed into an elongated sheet-like shape. For example, base material **23** is made of a synthetic resin such as polyethylene (PE) or polyethylene terephthalate (PET).

[0032] Base material **23** has an outer surface and an inner surface. First electrode **21** is arranged on the outer surface with an adhesive layer (not illustrated) interposed therebetween, and second electrode **22** is arranged on the inner surface with an adhesive layer (not illustrated) interposed therebetween. In other words, base material **23** is sandwiched between first electrode **21** and second electrode **22**. The inner surface is a surface holding second electrode **22** and facing rim **2**.

[0033] First electrode **21**, second electrode **22**, and base material **23** of grip detection device for steering wheel **3** are wrapped around rim **2**, and thus base material **23** is attached to rim **2**. Base material **23** is elongated (strip-shaped) such that rim **2** can be totally covered by wrapping base material **23** around rim **2** in the wrapping direction and also in a direction perpendicular to the wrapping direction when first electrode **21**, second electrode **22**, and base material **23** of grip detection device for steering wheel **3** are wrapped around rim **2**. In other words, base material **23** is formed in accordance with the shape and the size of rim **2**.

[0034] Second electrode **22** is a heater wire made of a conductive wire. Second electrode **22** is arranged around rim **2**. Specifically, second electrode **22** is arranged in a more inner layer of steering wheel **1** than first electrode **21**, i.e., closer to rim **2** than first electrode **21** is. One end of second electrode **22** and the other end of second electrode **22** are connected to control circuit **40**. Second electrode **22** is arranged on the inner surface of base material **23** in a winding (zig-zag)

manner. Specifically, second electrode **22** is, for example, a metal wire such as a copper wire, and is sewed onto the inner surface of base material **23** with sewing thread **24** to form into a winding pattern. Here, “second electrode **22** is sewed onto the inner surface of base material **23** with sewing thread **24**” means a state in which second electrode **22** is fastened to base material **23** by sewing thread **24** stitched on base material **23**.

[0035] It is to be noted that second electrode **22** according to the present embodiment is sewed onto the back of base material **23** with sewing thread **24**, but may be fixed to base material **23** using thermocompression bonding or the like. Moreover, second electrode **22** may be fixed to base material **23** using an adhesive.

[0036] Second electrode **22** is arranged on the surface of rim **2**, and is adhered to the surface of rim **2** with an adhesive (not illustrated). Second electrode **22** is sandwiched between rim **2** and base material **23**.

[0037] Control circuit **40** is embedded in, for example, a spoke. Control circuit **40** is electrically connected to first electrode **21** and second electrode **22**, and detects a touch on steering wheel **1** based on a signal transmitted from first electrode when user's hand comes close to steering wheel **1**. In other words, control circuit **40** detects whether user's hand is touching rim **2**.

[0038] Control circuit **40** is electrically connected to power supply circuit **40c**. By controlling power supply circuit **40c**, control circuit **40** supplies power from battery **39** to first electrode **21** and second electrode **22** in detecting whether the user's hand is touching rim **2**, or supplies power from battery **39** to second electrode **22** to generate the heat of second electrode **22**. In other words, power supply circuit **40c** is controlled by control circuit **40** to apply a measurement voltage to first electrode **21** and second electrode **22**. Moreover, power supply circuit **40c** is controlled by control circuit **40** to pass direct current through second electrode **22**. Second electrode **22** functions as a heater by the direct current passing through second electrode **22**. In this case, the direct current just passes through second electrode **22**, and thus as illustrated in FIG. **3** described later, one end of second electrode **22** and the other end of second electrode **22** are electrically connected to battery terminal **39a** and ground **38**, respectively. It is to be noted that second electrode **22** is not limited to the configuration including the heating function. Second electrode **22** may have a configuration of only floating from ground **38** without passing the direct current through second electrode **22**.

[0039] Next, the circuit configuration and the function of grip detection device for steering wheel **3** are described with reference to FIG. **3** and FIG. **4**.

[0040] FIG. **3** is the circuit configuration of grip detection device for steering wheel **3** according to Embodiment **1**. FIG. **4** is a diagram illustrating capacitances in cases where a first mode, a second mode, and a third mode are performed when user's hand is touching steering wheel **1**. Part (a) of FIG. **4** illustrates capacitance $C_{\text{sub.fgr}}$ and capacitance $C_{\text{sub.strHOD}}$ measured when user's hand is touching steering wheel **1** in the first mode. Part (b) of FIG. **4** illustrates capacitance $C_{\text{sub.strTHD}}$ measured when user's hand is touching steering wheel **1** in the second mode. Part (c) of FIG. **4** illustrates no capacitance measured when user's hand is touching steering wheel **1** in the third mode.

[0041] As shown in FIG. **3**, control circuit **40** has the following modes: the first mode in which first electrode **21** serves as a sensor electrode and second electrode **22** serves as a shield electrode; and the second mode in which first electrode **21** serves as a shield electrode and second electrode **22** serves as a sensor electrode. Control circuit **40** allows switching between the first mode and the second mode. In this case, control circuit **40** can detect a grip on steering wheel **1** based on the obtained first-mode capacitance and second-mode capacitance.

[0042] Control circuit **40** further has the third mode in which second electrode **22** acts as a heater. Control circuit **40** also allows switching among the first mode, the second mode, and the third mode. In this case, control circuit **40** can generate the heat of second electrode **22** by supplying power to second electrode **22**, as well as detect a grip on steering wheel **1** based on the obtained first-mode capacitance and second-mode capacitance.

[0043] Control circuit **40** has capacitance sensor circuit **40a**.

[0044] Capacitance sensor circuit **40a** calculates the capacitance between first electrode **21** and user's hand to detect a touch of the user's hand on steering wheel **1**, thereby detecting a grip on steering wheel **1** with the user's hand.

[0045] Capacitance sensor circuit **40a** has sensor terminal **41** and shield terminal **42**.

[0046] Sensor terminal **41** can apply the measurement voltage to the electrode connected to sensor terminal **41**, and thus allows the connected electrode to serve as a sensor electrode. Specifically, sensor terminal **41** allows first electrode **21** to serve as the sensor electrode in the first mode and second electrode **22** to serve as the sensor electrode in the second mode.

[0047] Shield terminal **42** can apply a shield voltage with the same phase as the measurement voltage to the electrode connected to shield terminal **42**, and thus allows the connected electrode to serve as a shield electrode. Specifically, shield terminal **42** allows second electrode **22** to serve as the shield electrode in the first mode and first electrode **21** to serve as the shield electrode in the second mode.

[0048] Grip detection device for steering wheel **3** includes first switch **31**, second switch **32**, third switch **33**, fourth switch **34**, fifth switch **35**, and sixth switch **36**, as well as control circuit **40**, power supply circuit **40c**, first electrode **21**, and second electrode **22**.

[0049] First switch **31** is arranged between sensor terminal **41** and second electrode **22**.

Specifically, first switch **31** has: one end connected to sensor terminal **41** and one end of second switch **32**; and the other end connected to one end of sixth switch **36**, the other end of third switch **33**, and one end of second electrode **22**.

[0050] Second switch **32** is arranged between sensor terminal **41** and first electrode **21**.

Specifically, second switch **32** has: one end connected to sensor terminal **41** and one end of first switch **31**; and the other end connected to the other end of fourth switch **34** and one end of first electrode **21**.

[0051] Third switch **33** is arranged between shield terminal **42** and second electrode **22**.

Specifically, third switch **33** has: one end connected to shield terminal **42** and one end of fourth switch **34**; and the other end connected to the other end of first switch **31**, one end of sixth switch **36**, and one end of second electrode **22**.

[0052] Fourth switch **34** is arranged between shield terminal **42** and first electrode **21**. Specifically, fourth switch **34** has: one end connected to shield terminal **42** and one end of third switch **33**; and the other end connected to the other end of second switch **32** and one end of first electrode **21**.

[0053] Fifth switch **35** is arranged between battery terminal **39a** and the other end of second electrode **22**. Specifically, fifth switch **35** has: one end connected to the other end of second electrode **22**; and the other end connected to power supply circuit **40c** for supplying power from battery **39**.

[0054] Sixth switch **36** is arranged between ground **38** and one end of second electrode **22**.

Specifically, sixth switch **36** has: one end connected to the other end of first switch **31**, the other end of third switch **33**, and one end of second electrode **22**; and the other end connected to ground **38**.

[0055] Control circuit **40** further has mode switching circuit **40b**.

[0056] Mode switching circuit **40b** can control switching for each of first switch **31**, second switch **32**, third switch **33**, fourth switch **34**, fifth switch **35**, and sixth switch **36**. Accordingly, mode switching circuit **40b** can switch modes including the first mode and the second mode at predetermined intervals. Mode switching circuit **40b** also can switch modes including the first mode, the second mode, and the third mode at predetermined intervals.

[0057] In such a circuit configuration, in the first mode, mode switching circuit **40b** can turn on second switch **32** and third switch **33** and turn off the other switches (first switch **31**, fourth switch **34**, fifth switch **35**, and sixth switch **36**).

[0058] In the second mode, mode switching circuit **40b** can turn on first switch **31** and fourth

switch **34** and turn off the other switches (second switch **32**, third switch **33**, fifth switch **35**, and sixth switch **36**).

[0059] In the third mode, mode switching circuit **40b** can turn on fifth switch **35** and sixth switch **36** and turn off the other switches (first switch **31**, second switch **32**, third switch **33**, and fourth switch **34**).

[0060] Next, the operation of grip detection device for steering wheel **3** is described with reference to FIG. **3** through FIG. **5**.

[0061] FIG. **5** illustrates cases where each mode is driven in a time sharing manner. Part (a) of FIG. **5** illustrates a case of switching between the first mode and the second mode. Part (b) of FIG. **5** illustrates a case of switching among the first mode, the second mode, and the third mode. In parts (a) and (b) of FIG. **5**, a period during which the first mode is performed is indicated by positive diagonal hatching, a period during which the second mode is performed is indicated by negative diagonal hatching, and a period during which the third mode is performed is indicated by crosshatching.

[0062] Capacitance sensor circuit **40a** can obtain the first-mode capacitance detected by first electrode **21** serving as the sensor electrode in the first mode and the second-mode capacitance detected by second electrode **22** serving as the sensor electrode in the second mode, and detect a grip on steering wheel **1** with user's hand based on the first-mode capacitance and the second-mode capacitance.

[0063] As illustrated in part (a) of FIG. **5**, when the third mode is not performed, grip detection device for steering wheel **3** can repeatedly switch between the first mode and the second mode at predetermined intervals.

[0064] Specifically, as illustrated in FIG. **3** and FIG. **4**, mode switching circuit **40b** turns on second switch **32** and third switch **33** and turns off the other switches to drive control circuit **40** in the first mode. In the first mode, capacitance sensor circuit **40a** passes alternating current through first electrode **21**, i.e., applies the measurement voltage to first electrode **21**. In this mode, when user's hand touches steering cover **11** of rim **2**, the capacitance of first electrode **21** corresponding to the touched portion changes. Accordingly, sensor terminal **41** of capacitance sensor circuit **40a** measures the capacitance of first electrode **21** based on the value of the current passing through first electrode **21** (the measurement voltage). In the equivalent circuit, as illustrated in part (a) of FIG. **4**, capacitance $C_{\text{sub.fgr}}$ and capacitance $C_{\text{sub.strHOD}}$ are connected in parallel with respect to first electrode **21** connected to capacitance sensor circuit **40a**. The capacitance measured in the first mode (the first-mode capacitance) includes capacitance $C_{\text{sub.fgr}}$ between the hand and first electrode **21** and capacitance $C_{\text{sub.strHOD}}$ between first electrode **21** and core metal **2a**. Capacitance $C_{\text{sub.fgr}}$ indicates a sensor value between the hand and first electrode **21**. Capacitance $C_{\text{sub.strHOD}}$ indicates the parasitic capacitance within steering wheel **1**. It is to be noted that "HOD" is the abbreviation for Hands On Detection.

[0065] After a predetermined period sufficient to measure the capacitance in the first mode has elapsed, mode switching circuit **40b** switches the mode from the first mode to the second mode. Specifically, mode switching circuit **40b** turns on first switch **31** and fourth switch **34** and turns off the other switches to drive control circuit **40** in the second mode. In the second mode, capacitance sensor circuit **40a** passes alternating current through second electrode **22**, i.e., applies the measurement voltage to second electrode **22**. In this mode, even when user's hand is touching steering cover **11**, the capacitance between the user's hand and first electrode **21** cannot be measured since second switch **32** is off, and the capacitance between the user's hand and second electrode **22** also cannot be measured since first electrode **21** intervenes between the user's hand and second electrode **22** as the shield electrode. Sensor terminal **41** of capacitance sensor circuit **40a** measures the capacitance of second electrode **22** based on the value of current passing through second electrode **22** (the measurement value). The capacitance measured in the second mode (the second-mode capacitance) is capacitance $C_{\text{sub.strTHD}}$ between second electrode **22** and core

metal **2a**. Capacitance C.sub.strTHD indicates the parasitic capacitance within steering wheel **1**. It is to be noted that “THD” is the abbreviation for Temperature & Humidity Detection.

[0066] Capacitance C.sub.strHOD and capacitance C.sub.strTHD are coupled within steering wheel **1**, and thus vary depending on a change in external environment such as temperature and humidity. However, it is known that there is a correlation in the capacitance variation depending on the change in external environment. For example, the capacitance is 50 pF under conditions where the ambient temperature is 25 degrees C. and the humidity is 30%, but the capacitance is 150 pF under conditions where the ambient temperature is 40 degrees C. and the humidity is 95%. This property can be used to estimate and calculate capacitance C.sub.strHOD from capacitance C.sub.strTHD.

[0067] When a function of the correlation, i.e., capacitance C.sub.strHOD=f(capacitance C.sub.strTHD), is obtained in advance, capacitance sensor circuit **40a** can calculate C.sub.strHOD based on C.sub.strTHD measured in the second mode, using the above function representing the correlation. After calculating C.sub.strHOD, capacitance sensor circuit **40a** can calculate capacitance C.sub.fgr by subtracting calculated capacitance C.sub.strHOD from the first-mode capacitance measured in the first mode (capacitance C.sub.fgr+capacitance C.sub.strHOD). Based on calculated capacitance C.sub.fgr, capacitance sensor circuit **40a** can detect that user's hand has gripped steering wheel **1**.

[0068] It is to be noted that the period (the predetermined period) during which the first mode is performed and the period (the predetermined period) during which the second mode is performed are equal to each other. For example, the period during which the first mode is performed and the period during which the second mode is performed are each several milliseconds.

[0069] Accordingly, mode switching circuit **40b** switches modes including the first mode and the second mode at predetermined intervals, i.e., drives the modes in a time sharing manner, and thus grip detection device for steering wheel **3** can detect that user's hand has gripped steering wheel **1**.

[0070] As illustrated in part (b) of FIG. 5, in grip detection device for steering wheel **3**, mode switching circuit **40b** also can repeatedly switch modes including the first mode, the second mode, and the third mode at predetermined intervals. For example, mode switching circuit **40b** may perform the third mode after performing the first mode and the second mode.

[0071] Specifically, when a predetermined period has elapsed after performing the second mode, mode switching circuit **40b** switches the mode from the second mode to the third mode.

Specifically, mode switching circuit **40b** turns on fifth switch **35** and sixth switch **36** and turns off the other switches to drive control circuit **40** in the third mode. In the third mode, capacitance sensor circuit **40a** passes direct current through second electrode **22** to cause second electrode **22** to function as a heater. In other words, second electrode **22** is electrically connected to battery **39** that supplies power to one end of second electrode **22**, thereby acting as a heater. In this case, sixth switch **36** is turned on, and thus the direct current flows into ground **38** after passing through second electrode **22**. In this mode, even when user's hand is touching steering cover **11**, the capacitance between the user's hand and first electrode **21** cannot be measured since second switch **32** is off. The capacitance between second electrode **22** and core metal **2a** also cannot be measured since first switch **31** is off.

[0072] In a period during which the third mode is performed, grip detection device for steering wheel **3** can heat steering wheel **1** by generating the heat of second electrode **22**.

[0073] The period during which the third mode is performed is longer than the period during which the first mode is performed and the period during which the second mode is performed. For example, the period during which the third mode is performed is at least several tens of milliseconds. Accordingly, it is possible to detect that user's hand has gripped steering wheel **1**, without losing the ability to increase temperature and the ability to keep warmth of steering wheel **1**.

Advantageous Effects

[0074] Next, the advantageous effects of grip detection device for steering wheel **3** according to the present embodiment are described.

[0075] For example, in conventional techniques, in the case where a heater pattern whose projected area is smaller than that of a sensor electrode is used as a shield electrode, when the heater pattern is arranged parallel to the sensor electrode, the parasitic capacitance between the sensor electrode and a core metal (a component for the GND potential) that is present inside a steering wheel is increased. In sensing a grip on the steering wheel, a signal in which the parasitic capacitance is added to a sensor value of the grip sensing is detected. The parasitic capacitance always varies depending on the external environment such as temperature and humidity, and thus the detected signal value also varies. Accordingly, this variation causes false sensing results.

[0076] In view of this, as described above, grip detection device for steering wheel **3** according to Technique **1** of the present embodiment includes: first electrode **21** arranged at rim **2** of steering wheel **1**; second electrode **22** arranged at rim **2** and in a more inner layer of steering wheel **1** than first electrode **21**; and control circuit **40**, in which first electrode **21** is wider in projected area than second electrode **22**, control circuit **40** allows switching between: a first mode in which first electrode **21** serves as a sensor electrode and second electrode **22** serves as a shield electrode; and a second mode in which first electrode **21** serves as the shield electrode and second electrode **22** serves as the sensor electrode, and control circuit **40** further: obtains a first-mode capacitance detected by first electrode **21** serving as the sensor electrode in the first mode and a second-mode capacitance detected by second electrode **22** serving as the sensor electrode in the second mode; and detects a grip on steering wheel **1** based on the first-mode capacitance and the second-mode capacitance.

[0077] With this, capacitance $C_{\text{sub.strHOD}}$ can be calculated based on capacitance $C_{\text{sub.strTHD}}$ indicated by the second-mode capacitance, and thus capacitance sensor circuit **40a** can calculate capacitance $C_{\text{sub.fgr}}$ by subtracting calculated capacitance $C_{\text{sub.strHOD}}$ from the first-mode capacitance (capacitance $C_{\text{sub.fgr}}$ +capacitance $C_{\text{sub.strHOD}}$).

[0078] Therefore, according to the present disclosure, it is possible to prevent the reduction in robustness of grip sensing and to achieve highly-accurate grip sensing.

[0079] In particular, in the conventional grip detection device for steering wheel, the third electrode serving as the shield electrode is further arranged between the first electrode which serves as the sensor electrode and the second electrode which performs the heating function. However, grip detection device for steering wheel **3** according to the present embodiment is configured to detect a grip on steering wheel **1** by switching between the first mode and the second mode, and thus it is not necessary to further arrange the third electrode serving as the shield electrode in the steering wheel like the conventional grip detection device for steering wheel. Accordingly, grip detection device for steering wheel **3** according to the present disclosure can prevent an increase in the number of components and also prevent an increase in manufacturing cost of grip detection device for steering wheel **3**.

[0080] Moreover, grip detection device for steering wheel **3** according to Technique **2** of the present embodiment is grip detection device for steering wheel **3** according to Technique **1**, in which control circuit **40** includes capacitance sensor circuit **40a** having: sensor terminal **41** that allows first electrode **21** to serve as the sensor electrode in the first mode and second electrode **22** to serve as the sensor electrode in the second mode; and shield terminal **42** that allows second electrode **22** to serve as the shield electrode in the first mode and first electrode **21** to serve as the shield electrode in the second mode, the grip detection device further includes: first switch **31** arranged between sensor terminal **41** and second electrode **22**; second switch **32** arranged between sensor terminal **41** and first electrode **21**; third switch **33** arranged between shield terminal **42** and second electrode **22**; and fourth switch **34** arranged between shield terminal **42** and first electrode **21**, and control circuit **40**: turns on second switch **32** and third switch **33** and turns off remaining switches in the first mode; and turns on first switch **31** and fourth switch **34** and turns off remaining

switches in the second mode.

[0081] With this, it is possible to switch between the first mode and the second mode by controlling first switch **31**, second switch **32**, third switch **33**, and fourth switch **34**. Accordingly, it is possible to detect a grip on steering wheel **1** based on the first-mode capacitance and the second-mode capacitance.

[0082] Moreover, grip detection device for steering wheel **3** according to Technique **3** of the present embodiment is grip detection device for steering wheel **3** according to Technique **1** or **2**, in which control circuit **40** further allows switching to a third mode in which second electrode **22** acts as a heater.

[0083] With this, second electrode **22** can act as a heater by performing the third mode.

Accordingly, it is possible to simultaneously implement, using second electrode **22**, the heating function of heating steering wheel **1** and the function of detecting a grip on steering wheel **1** with user's hand.

[0084] Moreover, grip detection device for steering wheel **3** according to Technique **4** of the present embodiment is grip detection device for steering wheel **3** according to Technique **3**, in which second electrode **22**, when acting as the heater, is connected to battery **39** that supplies power to a first end of second electrode **22**, a second end of second electrode **22** is connectable to ground **38**, the grip detection device further includes: fifth switch **35** arranged between battery **39** and the first end of second electrode **22**; and sixth switch **36** arranged between ground **38** and the second end of second electrode **22**, and control circuit **40** turns on fifth switch **35** and sixth switch **36** and turns off remaining switches in the third mode.

[0085] With this, in the third mode, power from battery **39** can be supplied to second electrode **22**, and thus it is possible to generate the heat of second electrode **22** to fulfill the heating function of heating steering wheel **1**.

[0086] Moreover, grip detection device for steering wheel **3** according to Technique **5** of the present embodiment is grip detection device for steering wheel **3** according to any one of Techniques **1** to **4**, in which control circuit **40** includes mode switching circuit **40b** that switches modes including the first mode and the second mode at predetermined intervals.

[0087] With this, it is possible to accurately detect the case where user's hand has gripped steering wheel **1** and the case where user's hand is away from steering wheel **1**, by switching between the first mode and the second mode at predetermined intervals.

[0088] Moreover, grip detection device for steering wheel **3** according to Technique **6** of the present embodiment is grip detection device for steering wheel **3** according to Technique **3** or **4**, in which control circuit **40** includes mode switching circuit **40b** that switches modes including the first mode, the second mode, and the third mode at predetermined intervals.

[0089] With this, it is possible to accurately detect, while heating steering wheel **1** using second electrode **22**, the case where user's hand has gripped steering wheel **1** and the case where user's hand is away from steering wheel **1**, by switching among the first mode, the second mode, and the third mode at predetermined intervals.

Other Embodiments

[0090] Although the grip detection device for steering wheel according to the present disclosure has been described based on the above embodiment, the present disclosure is not limited to the embodiment. Embodiments achieved by applying various modifications to the above embodiment conceived by a person skilled in the art without departing from the spirit of the present disclosure are also included in the present disclosure.

[0091] For example, grip detection device for steering wheel **3** according to the present disclosure may include multiple first electrodes **21** or multiple second electrodes **22**. Moreover, grip detection device for steering wheel **3** may include one first electrode **21** and multiple second electrodes **22**, or multiple first electrodes **21** and one second electrode **22**. Also in these cases, control circuit **40** may switch modes including the first mode and the second mode at predetermined intervals, or modes

including the first mode, the second mode, and the third mode at predetermine intervals.

[0092] Moreover, multiple grip detection devices for steering wheel **3** according to the present disclosure may be independently arranged in steering wheel **1**. For example, when steering wheel **1** is in the right position (the steering angle is 0°), grip detection device for steering wheel **3** may be arranged on each of the left side and the right side. It should be appreciated that three or more grip detection devices for steering wheel **3** may be arranged in steering wheel **1**.

[0093] Moreover, in grip detection device for steering wheel **3** according to the present disclosure, capacitance sensor circuit **40a** and mode switching circuit **40b** are included in one control circuit **40**, but capacitance sensor circuit **40a** and mode switching circuit **40b** may be each a separate independent circuit.

[0094] Those skilled in the art will readily appreciate that many modifications are possible in the above Embodiment without materially departing from the novel teachings and technical advantages of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the present disclosure.

Further Information About Technical Background to this Application

[0095] The disclosure of the following patent application including specification, drawings, and claims is incorporated herein by reference in its entirety: Japanese Patent Application No. 2024-030176 filed on Feb. 29, 2024.

Industrial Applicability

[0096] The grip detection device for the steering wheel according to the present disclosure is applicable to, for example, vehicle steering wheels.

Claims

1. A grip detection device for a steering wheel, the grip detection device comprising: a first electrode arranged at a rim of the steering wheel; a second electrode arranged at the rim and in a more inner layer of the steering wheel than the first electrode; and a control circuit, wherein the first electrode is wider in projected area than the second electrode, the control circuit allows switching between: a first mode in which the first electrode serves as a sensor electrode and the second electrode serves as a shield electrode; and a second mode in which the first electrode serves as the shield electrode and the second electrode serves as the sensor electrode, and the control circuit further: obtains a first-mode capacitance detected by the first electrode serving as the sensor electrode in the first mode and a second-mode capacitance detected by the second electrode serving as the sensor electrode in the second mode; and detects a grip on the steering wheel based on the first-mode capacitance and the second-mode capacitance.
2. The grip detection device according to claim 1, wherein the control circuit includes a capacitance sensor circuit having: a sensor terminal that allows the first electrode to serve as the sensor electrode in the first mode and the second electrode to serve as the sensor electrode in the second mode; and a shield terminal that allows the second electrode to serve as the shield electrode in the first mode and the first electrode to serve as the shield electrode in the second mode, the grip detection device further includes: a first switch arranged between the sensor terminal and the second electrode; a second switch arranged between the sensor terminal and the first electrode; a third switch arranged between the shield terminal and the second electrode; and a fourth switch arranged between the shield terminal and the first electrode, and the control circuit: turns on the second switch and the third switch and turns off remaining switches in the first mode; and turns on the first switch and the fourth switch and turns off remaining switches in the second mode.
3. The grip detection device according to claim 1, wherein the control circuit further allows switching to a third mode in which the second electrode acts as a heater.
4. The grip detection device according to claim 3, wherein the second electrode, when acting as the heater, is connected to a battery that supplies power to a first end of the second electrode, a second

end of the second electrode is connectable to ground, the grip detection device further includes: a fifth switch arranged between the battery and the first end of the second electrode; and a sixth switch arranged between the ground and the second end of the second electrode, and the control circuit turns on the fifth switch and the sixth switch and turns off remaining switches in the third mode.

5. The grip detection device according to claim 1, wherein the control circuit includes a mode switching circuit that switches modes at predetermined intervals, the modes including the first mode and the second mode.

6. The grip detection device according to claim 3, wherein the control circuit includes a mode switching circuit that switches modes at predetermined intervals, the modes including the first mode, the second mode, and the third mode.

7. The grip detection device according to claim 2, wherein the control circuit further allows switching to a third mode in which the second electrode acts as a heater.

8. The grip detection device according to claim 7, wherein the second electrode, when acting as the heater, is connected to a battery that supplies power to a first end of the second electrode, a second end of the second electrode is connectable to ground, the grip detection device further includes: a fifth switch arranged between the battery and the first end of the second electrode; and a sixth switch arranged between the ground and the second end of the second electrode, and the control circuit turns on the fifth switch and the sixth switch and turns off remaining switches in the third mode.
